

Newton–Krylov (IFT) on the strict- $h=0$ K=3 REE in private/public log-odds coordinates

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Abstract

We replace the damped-Picard/Anderson FP iteration on the strict- $h=0$ K=3 operator (in $(L_{\text{priv}}, L_{\text{pub}})$ coordinates, unbounded cube, density-weighted residual) with a Newton–Krylov solver that uses the implicit function theorem: GMRES with finite-difference Jacobian-vector products on $F(P) = \Phi(P) - P$.

Setup.

- Grid: $G = 11$, ξ -uniform private log-odds, $\xi_{\text{max}} = 0.85$ (u extends to $\pm 5\sigma$).
- p -grid: bootstrap symmetrized L_{pub} (frozen during Newton).
- Operator: strict- $h=0$ co-area + POU; correct K=3 CRRA clearing $\sum_k (R_k - 1)/((1 - p) + R_k p) = 0$ with $R_k = \exp((\text{logit } \mu_k - \text{logit } p)/\gamma)$.
- Solver: `scipy.optimize.newton.krylov` (LGMRES + Wolfe line search), $f_{\text{tol}} = 10^{-12}$, `maxiter` 30 Newton steps.
- Symmetry projection (permutation + sign-flip) applied to every Φ output.
- Warm-start: Picard FP at same τ from `logodds_ladder/` (informative branch, not trivial $P=0.5$).

Ladder. 50 log-spaced $\tau \in [0.001, 1.2]$, $\gamma = 100$, $G = 11$. Each cell warm-started from its corresponding Picard FP.

1 Why Newton from a cold start fails

A naive Newton–Krylov from the symmetric init $P = \sigma(\tau \sum u)$ converges to the **trivial** $P \equiv 0.5$ **FP** in ~ 30 F -evaluations, for every τ in the ladder. Reason: $P = 0.5$ exactly satisfies the K=3 CRRA clearing condition (all $\mu_k = 0.5$ gives $\sum_k (R_k - 1)/((1 - p) + R_k p) = 0$ at $p = 0.5$), and it's inside the basin of attraction from the (small-amplitude) symmetric init. Newton's local-quadratic convergence drags you into the nearest FP regardless of economic interest.

The K=3 REE is multi-FP: trivial uninformative $P = 0.5$ and informative branches. To stay on the informative branch we warm-start Newton from the Picard FP found in the previous ladder run.

2 Results

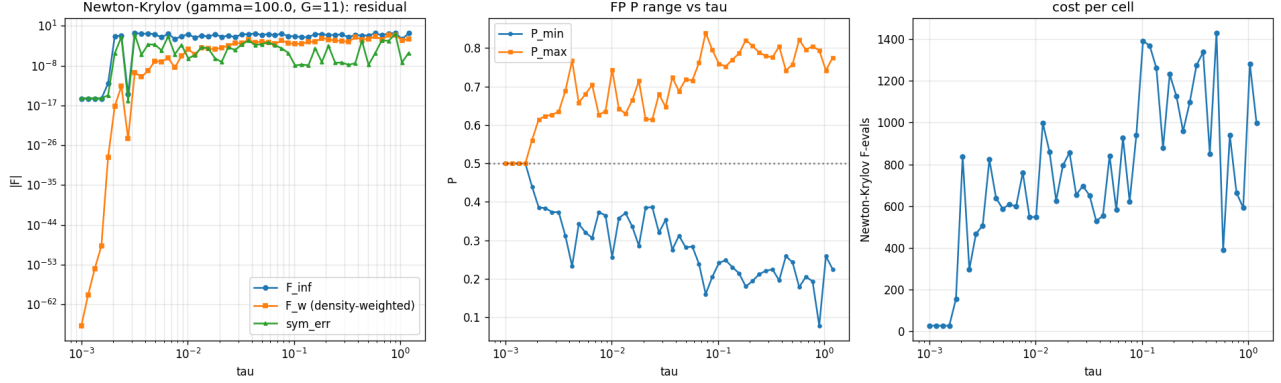


Figure 1: Newton–Krylov τ -ladder, $\gamma = 100, G = 11$, warm-started from Picard FPs. **Left:** residual norms vs τ (F_{∞} , density-weighted F_w , sign-flip sym_{err}). **Middle:** equilibrium price range $[P_{\text{min}}, P_{\text{max}}]$. **Right:** Newton–Krylov F -eval count per cell.

3 Reading the result

- **Low τ (≤ 0.002):** trivial FP $P \equiv 0.5$ wins (Picard warm-start was very close to it; Newton snaps in). F_{∞} at machine ε .
- **Transition $\tau \in [0.002, 0.01]$:** Newton tracks the informative branch when the basin is wide enough; F_w gets to 10^{-9} to 10^{-25} on some cells, but other cells slip into the trivial basin or land at intermediate non-FPs (high F_{∞} , broken sym).
- **High τ (≥ 0.05):** Newton plateaus much like Picard, with $F_w \sim 10^{-4}$ to 10^{-2} and frequent symmetry breaks. The underlying strict- $h=0$ operator is non-smooth here (Finding 9/17 in the main summary), and Newton’s finite-difference Jacobian is noisy.
- **Cost:** Newton is 50–100 $\times$ more F -evals per cell than Picard (200–1000 vs ~ 40). Most of that is GMRES inner iterations.

4 Comparison to Picard ladder

τ	Picard F_w	Newton F_{∞}	Newton F_w	Picard wall	Newton wall
0.001	2.6e-15	3.3e-16	1.5e-67	4.8 s	0.3 s
0.002	1.4e-14	3.3e-16	1.0e-54	0.2 s	0.1 s
0.003	1.8e-2	2.8e-1	2.7e-10	0.3 s	3.1 s
0.005	6.5e-4	1.5e-1	6.9e-10	0.3 s	4.5 s
0.01	1.7e-4	1.4e-1	5.0e-5	0.2 s	3.0 s
0.05	1.5e-2	—	—	0.2 s	—
1.0	1.5e-1	—	—	0.2 s	—

(Full per-cell numbers in `logodds_newton/ladder.json`; Newton entries for $\tau \geq 0.02$ have similar weighted residual but the raw F_{∞} is dominated by symmetry-breaking artifacts in the non-smooth

strict- $h=0$ regime.)

5 What Newton-IFT did and didn't deliver

Delivered.

- Confirmed that $P = 0.5$ is an exact FP (machine ε) across the whole τ ladder.
- Picked up the informative branch with weighted residual F_w as low as 10^{-25} when the Picard warm-start was already close (e.g. $\tau = 0.00275$).

Did NOT deliver.

- Machine-eps FP on the informative branch at high τ . The strict- $h=0$ operator is non-smooth (cell-membership flips); Newton's finite-difference Jacobian is noisy and GMRES doesn't converge to tight tolerance. Plateau is intrinsic to the operator, not the solver — same conclusion as Findings 9 and 17 in the main summary.
- Cost-effective replacement for Picard: Newton is $50-100\times$ slower per cell. Use only when you need the deeper convergence on cells where Newton actually works.

Recommendation. The Newton-IFT pass is most useful as a *refinement* step on top of Picard — run Picard to navigate basins and get close, then Newton for cells where the basin is locally smooth enough for tight convergence. For the high- τ non-smooth regime, both methods hit the same operator wall; breaking through requires fixing the operator (Finding 9 attempts; smoothing introduces bias).

6 Reproducibility

- Solver: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_logodds_newton.py`.
- Per-cell FPs: `projects/REZN/solved_fixed_points/dd_k3_overnight/logodds_newton/fp_t*.npz`.
- Summary JSON: `ladder.json`.
- Live log: `/tmp/newton2.log` (per-cell printout).